EXPERIMENT 6

6.1.1Incremented Date

Algorithm:

1. **Start**
2. **Input** date, month, year.
3. Set a variable flag = True.
4. **Validate month and year**

   If year < 0 OR month < 0 OR month > 12
   → set flag = False.
5. **Find maximum days of the month**

   If month is **1,3,5,7,8,10,12** → maxdays = 31

   Else if month is **4,6,9,11** → maxdays = 30

   Else (February):

   - If year is divisible by 4 → maxdays = 29
   - Else → maxdays = 28
6. **Check date validity**

   If date > maxdays
   → flag = False
7. **Calculate next date**

   date = date + 1
8. **If date exceeds maxdays**

   Set date = 1

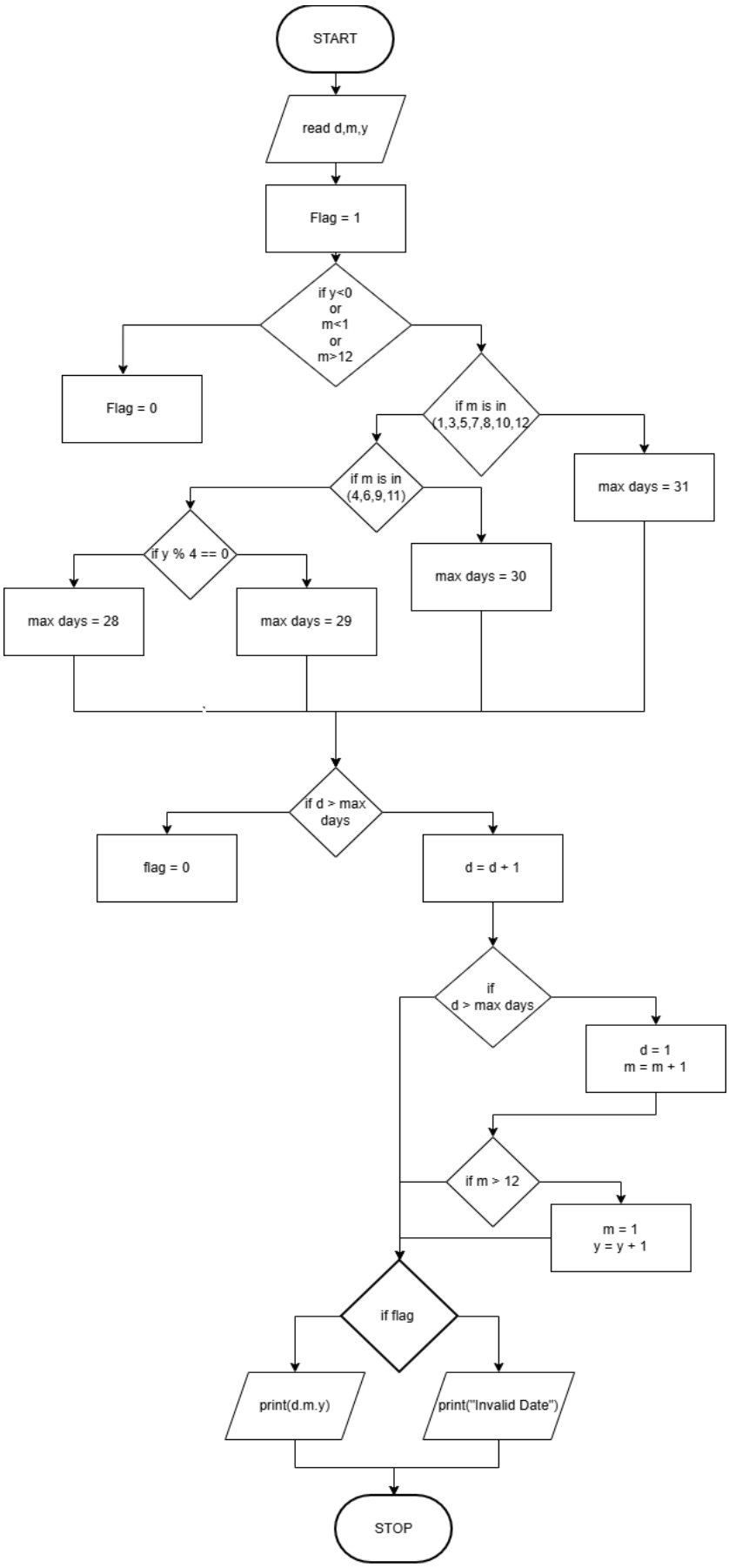
   increase month = month + 1
9. **If month exceeds 12**

   Set month = 1

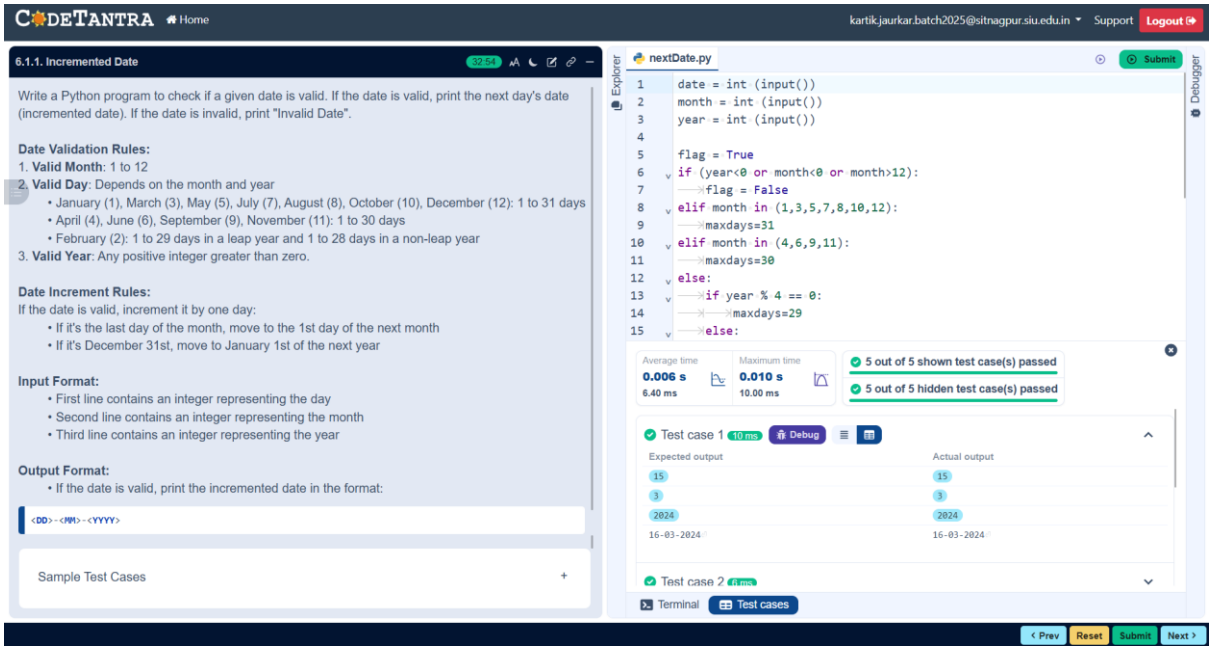
   Increase year = year + 1
10. **Display result**

    - If flag is True → print next date in format **dd-mm-yyyy**
    - Else → print **Invalid Date**
11. **Stop**

Flowchart:


START
read d,m,y
Flag = 1
if y<0
or
m<1
or
m>12
Flag = 0
if m is in
(1,3,5,7,8,10,12
max days = 31
if m is in
(4,6,9,11)
max days = 30
if y % 4 == 0
max days = 28
max days = 29
if d > max
days
flag = 0
d = d + 1
if
d > max days
d = 1
m = m + 1
if m > 12
m = 1
y = y + 1
if flag
print(d.m.y)
print("Invalid Date")
STOP

Execution: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6.1.1. Incremented Date
32.54
Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".
Date Validation Rules:
1. Valid Month: 1 to 12
2. Valid Day: Depends on the month and year
• January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
• April (4), June (6), September (9), November (11): 1 to 30 days
• February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. Valid Year: Any positive integer greater than zero.
Date Increment Rules:
If the date is valid, increment it by one day:
• If it's the last day of the month, move to the 1st day of the next month
• If it's December 31st, move to January 1st of the next year
Input Format:
• First line contains an integer representing the day
• Second line contains an integer representing the month
• Third line contains an integer representing the year
Output Format:
• If the date is valid, print the incremented date in the format:
<DD>-<MM>-<YYYY>
Sample Test Cases
Explorer
nextDate.py
Submit
Debugger
date = int (input())
month = int (input())
year = int (input())
flag = True
if (year<0 or month<0 or month>12):
flag = False
elif month in (1,3,5,7,8,10,12):
maxdays=31
elif month in (4,6,9,11):
maxdays=30
else:
if year % 4 == 0:
maxdays=29
else:
Average time
0.006 s
6.40 ms
Maximum time
0.010 s
10.00 ms
5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed
Test case 1
10 ms
Debug
Expected output
Actual output
15
15
3
3
2024
2024
16-03-2024
16-03-2024
Test case 2
Terminal
Test cases
Prev
Reset
Submit
Next

6.1.2. Factorial of a Number:

Algorithm:

Step 1: Start

Step 2: Input an integer n

Step 3: Initialize a variable fact = 1

Step 4: Repeat a loop from i = 1 to n

Step 5: n each iteration:

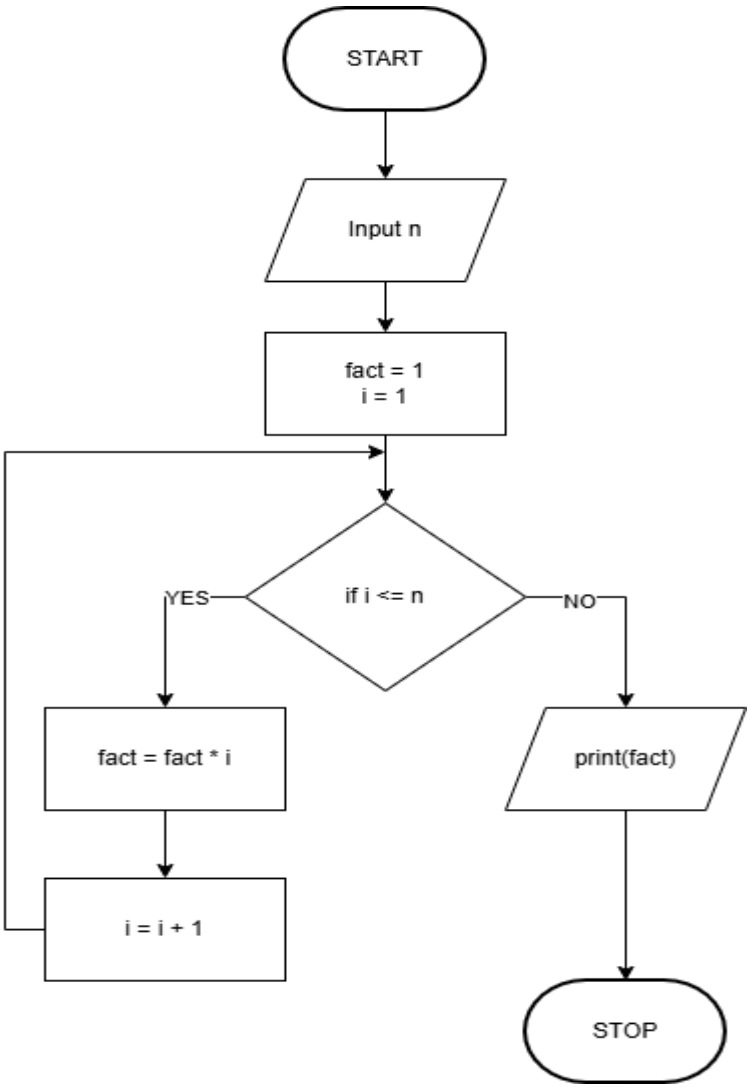
- Update fact = fact × i

Step 6: End loop

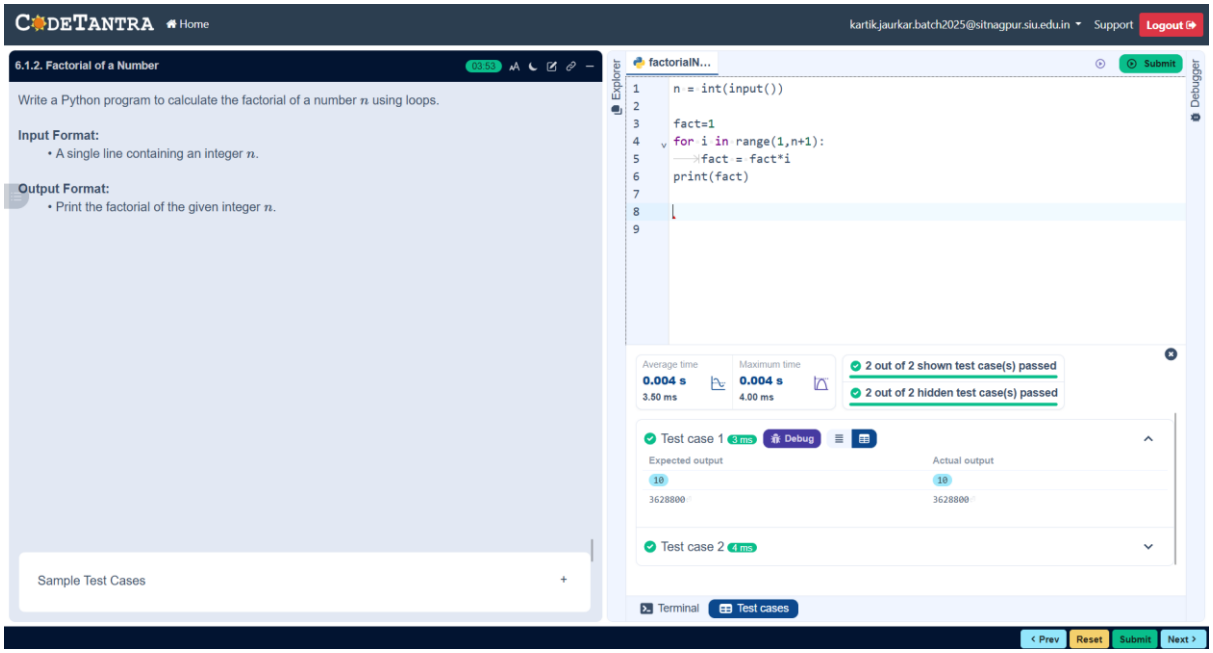
Step 7: Display the value of fact

Step 8: Stop

Flowchart:



Execution: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6.1.2. Factorial of a Number
03:53
Write a Python program to calculate the factorial of a number n using loops.
Input Format:
• A single line containing an integer n.
Output Format:
• Print the factorial of the given integer n.
Sample Test Cases
factorialN...
Submit
Explorer
Debugger
n = int(input())
fact=1
for i in range(1,n+1):
fact = fact*i
print(fact)
Average time
0.004 s
3.50 ms
Maximum time
0.004 s
4.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
3 ms
Debug
Expected output
10
3628800
Actual output
10
3628800
Test case 2
4 ms
Terminal
Test cases
Prev
Reset
Submit
Next